Visualising Emotion

How far does an automatic reading of an AI drawing agree with an automatic reading of the words that made it?

A complete explanation of the project

MSc Artificial Intelligence dissertation · Shre Hari Prem Anandh
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Every figure in this document was re-verified against the analysis on 9 September 2026

The project in one paragraph. At the Festival of the Mind 2024, members of the public typed a few words about how they were feeling, and DALL·E drew each entry as a black-and-white line drawing — 1,025 records; the number of contributors is unknown. This project builds and validates a way to measure how far an automatic reading of the drawing agrees with an automatic reading of the words. It finds that agreement is real and well above chance where the describer reports a readable face — and, once the description stops inventing an expression, at chance where it reports a scene — that the same words do not reliably give the same reading, and that the method used to measure it matters a great deal.

1 The question

You can already check whether an AI drew what a prompt asked for, provided the prompt names objects. Ask for *a red car* and a metric can look for a car and check that it is red. That is what most text-to-image evaluation does.

Feelings do not work that way. If somebody types *tired but happy*, there is no object to look for. There is no agreed answer key. And the feeling is mixed, which most emotion tools handle badly.

The precise question this project answers:

To what extent does an automatic image-emotion reader agree with prompt-derived emotion labels for AI-generated monochrome drawings, and how does that agreement vary by visual content and by repeated prompts?

That wording matters, and it is deliberately narrower than *does the drawing convey the feeling*. This project measures **agreement between two automatic systems** — one reading the words, one reading the drawing. It does not establish that a human viewer perceives the same feeling. That distinction is carried honestly through everything that follows.

2 Why the question is open

97 papers were collected, read and triaged; 24 were excluded and 73 kept as core evidence, organised into eight themes. The literature establishes four things that, taken together, leave a gap.

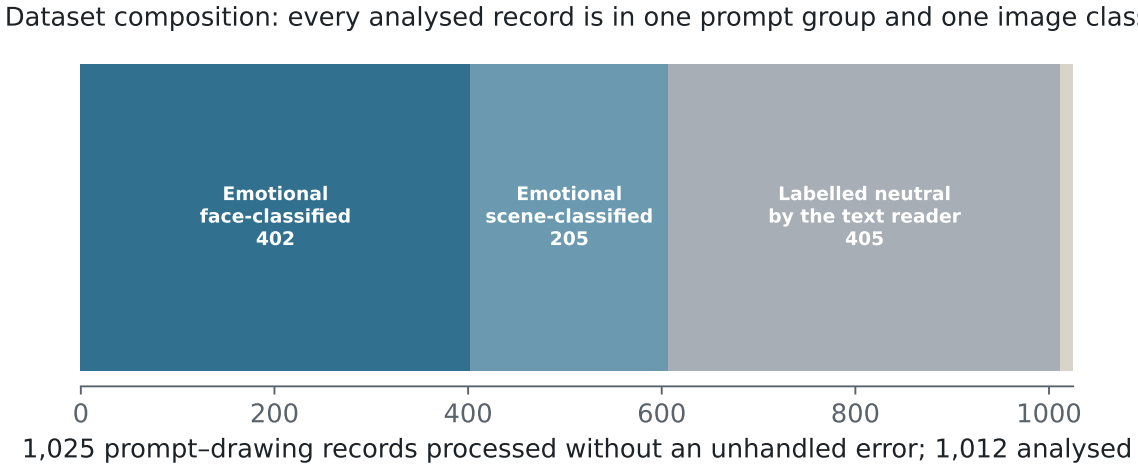
- **Emotion in text is well studied.** GoEmotions supplies a 28-label taxonomy (27 emotions plus neutral), a large labelled corpus and an off-the-shelf classifier. But short, first-person self-report is the hardest register for these tools, and free self-report resolves roughly 27 blended categories rather than a tidy handful.

- **Emotional image *generation* is well studied.** Systems exist that generate images towards a chosen target emotion. Alvarado (2025) finds that in text-to-image output lightness and chroma, rather than hue, are the principal colour carriers of emotional meaning; that study models colour features only, so it gives no evidence either way about whether form and line can do the same.
- **Text-to-image *evaluation* is well studied, but for objects.** The metrics score whether the picture contains what the prompt named. Recent work shows these metrics are fragile enough to be flipped by small perturbations.
- **Reading emotion back out of an image is possible but weak.** Vision-language models are positively biased and order-sensitive readers of emotion.

The gap. Within the 73-paper corpus reviewed here, no work measures prompt-reader / image-reader agreement for short, unmediated, mixed-affect first-person self-reports rendered as monochrome line art, nor the stability of that reading under repeated prompts. This dataset is unusually well suited to asking: the prompts are real, first-person, short and often mixed; and because the drawings are monochrome line art, the channel the literature says usually carries emotion — colour — is simply absent.

An important restraint. The colour literature motivates why this dataset is a hard case. It is *not* a result of this project. There is no colour-control condition here, so no claim is made about the effect of removing colour. Every finding applies within this monochrome corpus.

3 The dataset, and what it took to use it



1,025 prompt–drawing records. Median prompt length is about four words. Visitors were invited to describe how they were feeling but the interface accepted free text, so a substantial minority of entries are object or scene requests rather than self-report. Records are not linked to individuals and 249 repeat an earlier prompt, so the number of contributors is unknown and no greater than 763. Before any analysis, three properties had to be handled.

- **Three incompatible filename schemes** in one folder: `img_00001.PNG` for ids 1–199, `img_200.PNG` for 200–999, `img_t1000.PNG` for 1000 and above — and two files use a lowercase extension. A naive loader resolves under 20% of the corpus; the loader written here resolves 100%.
- **Hidden provenance metadata.** Every PNG carries a C2PA/JUMBF manifest containing an RFC-3161 signing timestamp. Recovering it gave a creation time per image and revealed that **records 1–13 were generated before the festival** — developer tests. They are processed but excluded from analysis, so nothing is silently dropped.

- **Repeated prompts.** 87 distinct prompts were drawn more than once, covering 336 records (43 of the groups are pairs). This looked like a nuisance and became a finding.

Dataset accounting	**Records**
Processed by the pipeline (no unhandled errors)	1,025
Pre-festival developer tests, excluded	13
Analysed	**1,012**
prompts the text reader labels emotional	607
prompts the text reader labels neutral (the specificity set)	405
face-classified (describer reports a person and an expression)	591
scene-classified	421
Distinct prompts (identity from pre-redaction text)	763
Records repeating an earlier prompt	249

4 The method

The supervisor's instruction was decisive: **never ask a vision model what emotion a picture shows.** Describe the picture in plain factual terms, then infer the feeling from that description with a separate model. The pipeline implements exactly that.

4.1 Stage by stage

1. **Load and redact.** Prompts are loaded through a single loader that applies name redaction, so the analysis, the figures and the demo app cannot diverge.
2. **Describe.** BLIP-VQA is asked five fixed questions about the drawing: *What is the facial expression?*, *What is the person doing?*, *What is the body posture?*, *How many people are there?*, *What is in the picture?* The answers are assembled into one neutral sentence. **The word "emotion" never appears in any question** — though the expression question returns a bare emotion word for 41% of records, and where it does the next stage follows it 99.8% of the time (see Limitations).
3. **Extract, image side.** A zero-shot natural-language-inference model reads that description against the hypothesis template *This scene conveys* {}. across seven emotion families, returning a ranked distribution.
4. **Extract, text side.** GoEmotions reads the person's own words, and its 28 labels are collapsed onto the same seven families, so both sides live in one label space.
5. **Compare.** Four similarity measures, each referenced to chance.

4.2 Models used

Model	**Size**	**Role**
`Salesforce/blip-vqa-base`	∼385M	Describes the drawing (five questions)
`facebook/bart-large-mnli`	∼407M	Reads a feeling from that description
`SamLowe/roberta-base-go_emotions`	∼125M	Reads a feeling from the person's words
OpenCLIP ViT-B/32 (LAION-2B)	∼151M	*Frozen baseline only* — the older "ask CLIP directly" method, kept for comparison

4.3 The four similarity measures

Similarity, not matching — *happy* and *excited* should count as close, not as a miss.

1. **Valence–arousal similarity.** Each side's ranked emotions are collapsed to a single point on a pleasantness/energy plane using fixed centroid coordinates per emotion, and the measure is one minus the normalised distance between the two points. *This is a designed proxy built on hand-set centroids, not a continuous lexicon rating.*
2. **Valence-sign agreement.** Do both sides land on the same side of pleasant/unpleasant?
3. **Emotion-family agreement.** Do both sides pick the same emotion family outright?
4. **Top-3 overlap.** How much of the two ranked distributions actually coincides?

5 What was tried and rejected

Each link was tested before being trusted. The failures shaped the design and are findings in their own right.

Approach	Why it was rejected (evidence)
GoEmotions on the image descriptions	Collapses to *neutral* above 0.95. It is trained on first-person feeling words; a third-person visual description contains none.
A single BLIP caption fed to the reader	Captions too terse (*a man with hands in the air*); the reader collapses to *surprise* for almost everything.
BART-MNLI on the *prompt* side	Over-reads *surprise* on short first-person text: 333 of 1,025 prompts came back as surprise. GoEmotions used instead.
Qwen2.5-VL-3B as the describer	7 GB in half precision exceeded a 16 GB laptop; thrashed into swap with **zero progress after 4 h 45 min**. Never produced a description.
Florence-2 as the describer	Hard library incompatibility (`forced_bos_token_id`) on transformers 5.14.
Asking CLIP directly for the emotion	Agreed far less with the prompt reader in the like-for-like comparison (see Findings), and manual inspection found clear errors: arms-raised celebration read as *anger* 0.73.
BLIP-VQA (chosen)	∼0.4B, roughly 1.5 s per record, and its "facial expression" answer supplies exactly the affective cue the next stage needs.

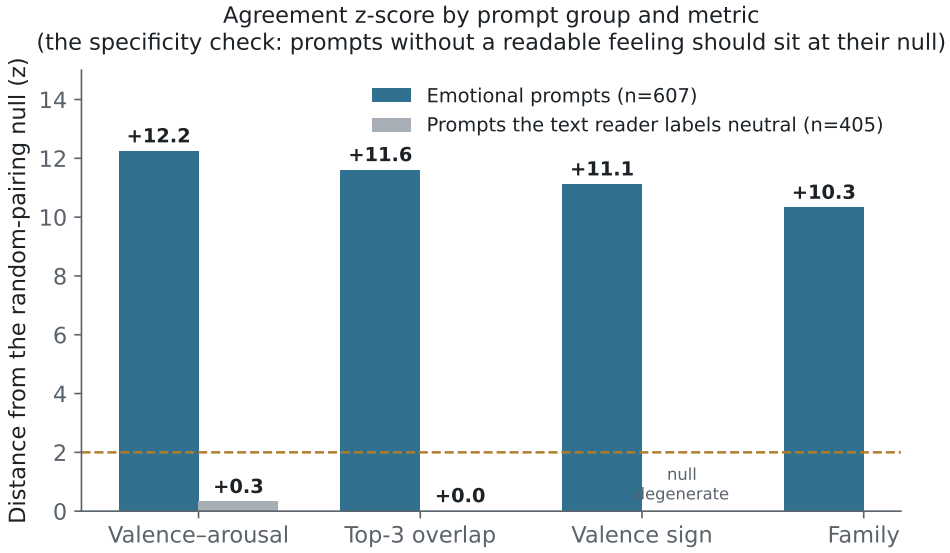
6 How the numbers are kept honest

This is the part that makes the findings usable, and it is where most of the methodological effort went.

6.1 A random-pairing null for every metric

Within each subset, the image side is shuffled and re-paired against the untouched prompt side, and all four metrics are recomputed. This is repeated **300 times** with a fixed seed. The average is the chance level; the spread gives a z-score. Every number reported in this project is quoted against its own null. A result counts only insofar as it beats chance.

6.2 A specificity check



405 of the 1,012 prompts are labelled neutral by the text reader — *pigs*, *mickey mouse*, *disney characters*. If the method were finding agreement where the prompt side has no readable feeling, it would score above chance on these. Only the valence–arousal measure is informative here: 76.6% against a null of 76.6%, $z = +0.3$. Valence-sign and family agreement are degenerate on this subset — every prompt shares the neutral label, so permutation cannot change them and their z is zero by construction, not by measurement. Two caveats: 68 of the 405 (16.8%) contain a plain affect word (calm, content, tired, relaxed, stress), and two consist of a name alone; on the cleaned 335 the z is $+0.04$, still at the null. The set is therefore better described as "prompts the text reader could not read affect from" than as prompts with no feeling.

Why this matters. An instrument that reported a signal where the prompt side has none could not be trusted anywhere. This check, resting on one measure, is one reason — not the only one — that the other numbers can be read.

6.3 A robustness check on repeated prompts

249 records repeat an earlier prompt, so treating all 1,012 as independent would overstate confidence. Every headline was re-run on the 763 distinct prompts (452 emotional). The overall and face results are essentially unchanged (z $+9.3$ to $+10.8$; face valence-sign 71.8%, z+8.4). The scene result weakens substantially (valence-sign 37.3%, z+3.5; family 30.8%, z+2.6, n falling from 205 to 133), and its family z does not survive a Holm–Bonferroni correction across the 36 non-degenerate z-scores (26 survive at family-wise 0.05).

6.4 Degenerate nulls are refused — automatically

Where a subset shares one label by construction — for example the neutral-labelled records — shuffling cannot change the score, so the "null" is an identity rather than a baseline. The analysis detects this from the null's own spread (zero standard deviation) and prints no z for such a measure: the sign measure on the single-valence subsets, the family measure on the positive-prompt subset, and both on the neutral-labelled subset. The negative-prompt family null is not degenerate.

6.5 Confidence intervals, and a test of the difference

Every reported rate carries a 95% confidence interval from a clustered bootstrap (10,000 draws, resampling prompt groups rather than records, because 249 records repeat an earlier prompt). The face/scene difference is tested directly with a clustered permutation test (10,000 draws): $+30.9$ pp [$+22.6$, $+40.6$] for valence sign and $+26.3$ pp [$+18.4$, $+35.3$] for family, $p = 0.0001$ for both. Cohen's κ for family agreement is 0.23 [0.18, 0.29] on the emotional prompts, 0.30 [0.23, 0.37] face-classified and 0.13 [0.06, 0.19] scene-classified — fair, fair and slight. After a Holm–Bonferroni correction across all 36 non-degenerate z-scores, 26 survive; the ten that do not are the control cells, the single-valence valence–arousal and overlap cells, and the distinct-prompt scene family result.

7 Chance is not zero

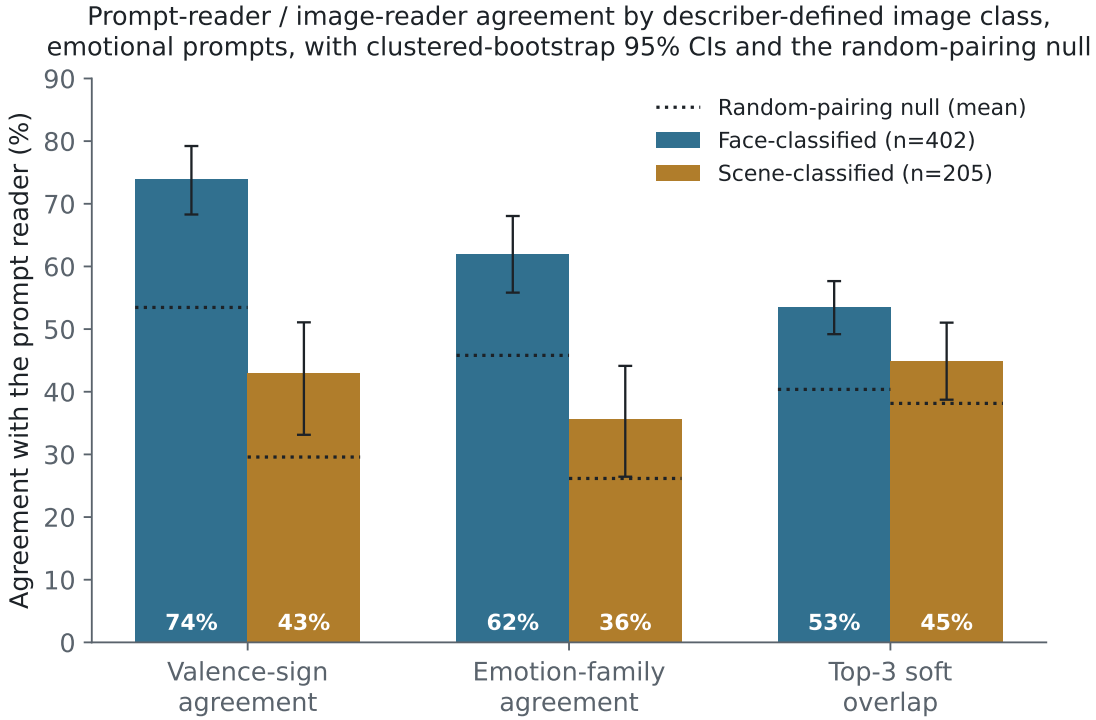
A late and important discovery, which changed how results are reported.

> The valence–arousal similarity measure **cannot fall below 41%** — that is its mathematical floor, given the fixed emotion coordinates. Two *randomly paired* records already score about **76–77%**, because most emotions sit close together on the plane.
>
> So a score of 71% looks like a pass and is in fact **below chance**. On a bare 0–100 scale that is invisible.

There is a second consequence. That measure returns roughly 84% for face drawings and 83% for scene drawings — it cannot tell them apart. The face/scene difference lives in valence-sign and emotion-family agreement. The reporting was changed accordingly: those two measures lead, and the valence–arousal score is always shown against its measured chance line.

8 Findings

8.1 Agreement is real, and much stronger where the describer reports a face

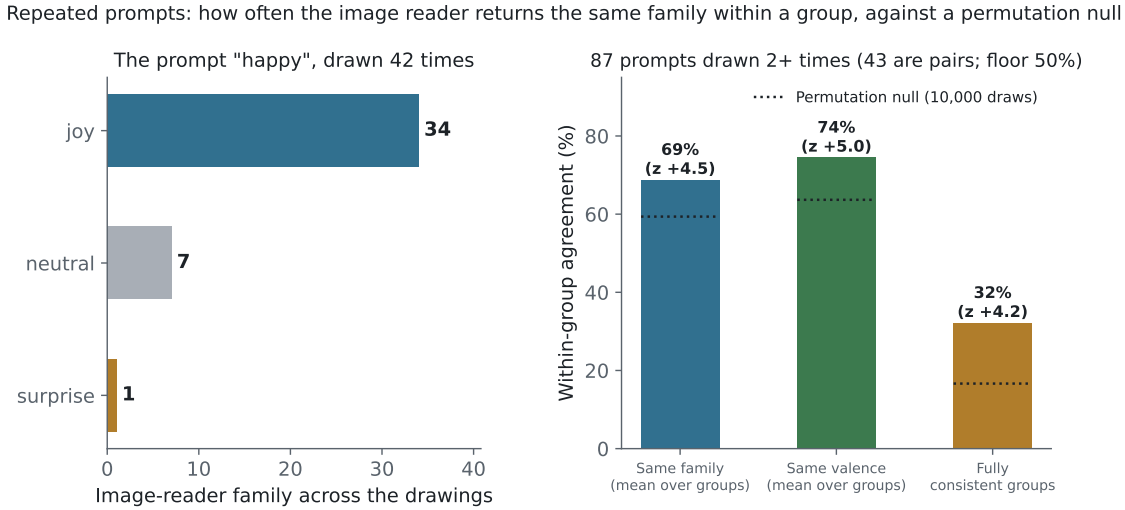


Emotional prompts only	Valence sign	Emotion family	Top-3 overlap
Face drawings ($n = 402$)	**73.9%** (z+10.4)	**61.9%** (z+10.3)	53.4% (z+11.6)
Scene drawings ($n = 205$)	42.9% (z+5.8)	35.6% (z+4.4)	44.9% (z+4.9)
All emotional ($n = 607$)	64.0% (z+11.1)	53.0% (z+10.3)	50.6% (z+11.6)

Both are above their nulls (95% intervals: face valence-sign [68.3, 79.2], scene [33.1, 51.1]). The difference is $+30.9$ pp [$+22.6$, $+40.6$] for valence sign and $+26.3$ pp [$+18.4$, $+35.3$] for family, clustered permutation $p = 0.0001$. No ratio is quoted, because the two nulls differ. On distinct prompts only the face result holds (71.8%, z+8.4) while the scene result weakens (37.3%, z+3.5).

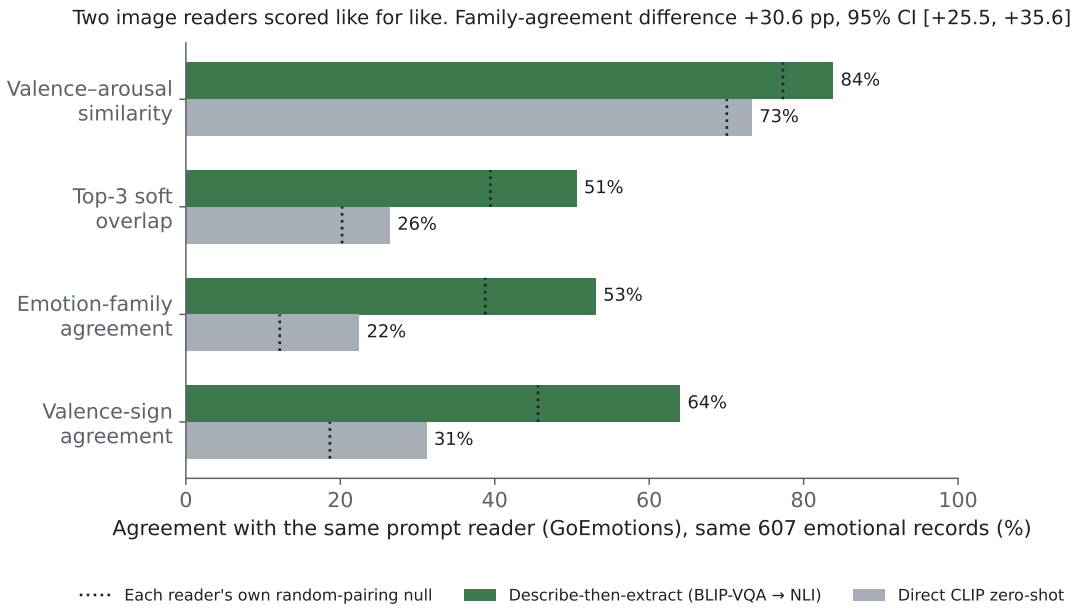
Stated carefully: the pipeline finds stronger prompt-reader / image-reader agreement for drawings it *classifies* as containing a readable face. It is not evidence about faces as such, because the classification and the emotion reading share a source — see Limitations. The template ablation below shows the scene rows were an artefact of that shared source: described honestly, scene-classified drawings give nothing above chance for this reader.

8.2 The same words do not reliably give the same reading



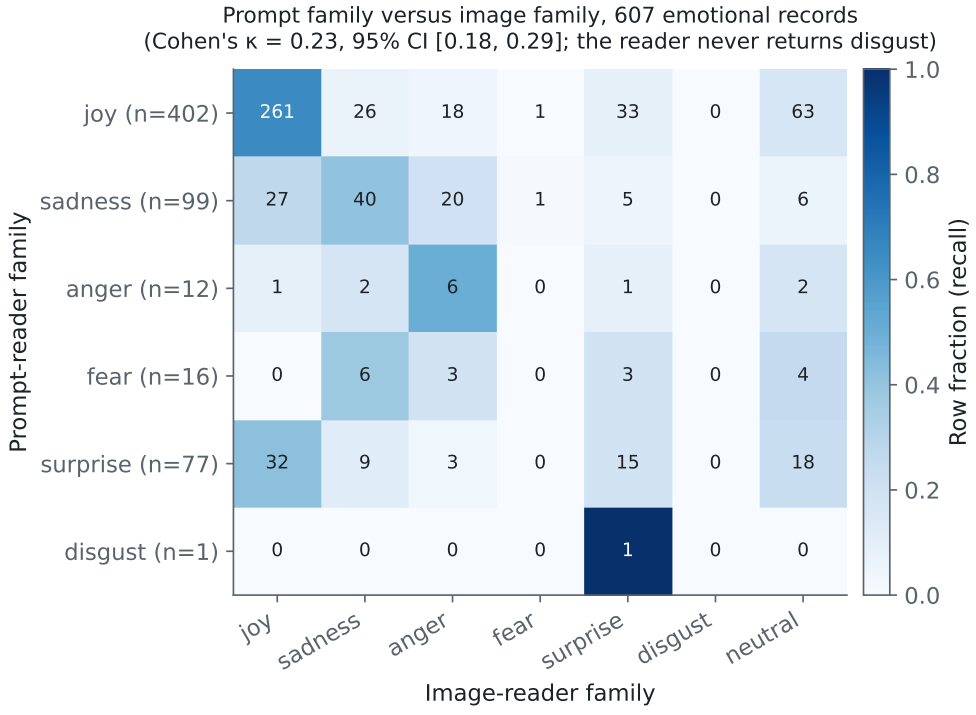
87 prompts were drawn between 2 and 42 times each, covering 336 records. Given *identical* words, the image reader returns the same family for **68.8%** of the drawings within a group, against a permutation null of 59.4% (z+4.5, $p = 0.0001$; 10,000 draws), and **28 of the 87** groups are fully consistent against 14.5 by chance (z+4.2). Half the groups are pairs, whose modal agreement cannot fall below 50%. The reading is more consistent than chance but far from fixed — and it measures the whole prompt→image→reader chain, not the generator alone, since generation parameters were never recorded and the reader has no reliability estimate of its own. A single drawing is one sample, not a fixed answer.

8.3 The method matters: describe-then-extract agrees far more closely



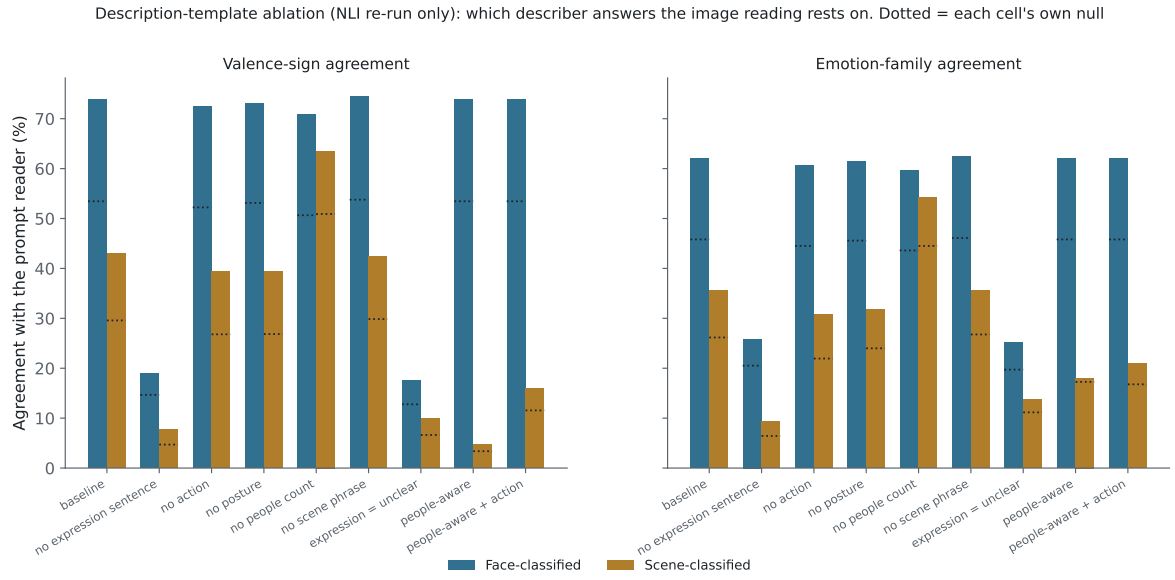
Like for like — same GoEmotions prompt reader, same 607 emotional records, same measures, only the image reader varied: describe-then-extract reaches **53.0%** emotion-family agreement against **22.4%** for asking CLIP directly ($+30.6$ pp; clustered 95% interval $[+25.5, +35.6]$; cluster-level $p < 0.0001$), 64.0% against 31.1% on valence sign, and 50.6% against 26.4% on top-3 overlap. Direct CLIP reads 283 of the 607 drawings as neutral. Neither reader has ground truth, so this ranks two image readers against a third model's labels: they disagree, and describe-then-extract is far closer to the text reader. The supervisor's redirection was supported rather than assumed. (The earlier figure of 49.3% against 25.3%, on 608 records with the NLI prompt reader, is kept only as the historical v1 record.)

8.4 Where the readers disagree



Three families are too rare on the prompt side to support family-level claims (fear $n = 16$, anger $n = 12$, disgust $n = 1$), and the image reader never returns disgust and returns fear four times across all 1,012 records. Every high-confidence disagreement (both readers ≥ 0.8, families differ: 38 records) was examined against the drawing and assigned a primary failure mode: 19 trace to the describer (a face or expression invented, or read wrongly), 5 to the image reader, 12 to the prompt side (ambiguous or mixed feelings), and 2 to a drawing that does not depict its prompt.

8.5 What the reading rests on: the template ablation



Each record's five cached describer answers were re-composed under an altered description template and only the NLI stage re-run (`ablate_nli.py`); a control re-run of the unmodified template reproduces every baseline number exactly (0 of 1,012 top-1 readings change), so every difference is a template effect. Removing the expression sentence collapses agreement — all-emotional family 53.0% $\rightarrow$ 20.3% (null 16.6%, $z{+}3.8$), valence sign 64.0% $\rightarrow$ 15.3% (11.4%, $z{+}5.1$); fixing the expression word to "unclear" gives the same collapse (family 21.3%, $z{+}3.9$), and the reader's output falls to surprise or neutral (514 and 368 of 1,012). Removing the action, posture or scene clause moves the rates by 1–3 pp. Removing the people count raises rates and nulls together (scene valence sign 63.5% vs 50.9%, $z{+}5.0$): the "0" token pulls scene readings toward neutral, and z barely moves.

Under a **people-aware template** — *A drawing showing {scene}.* wherever the describer reports no person, every other record unchanged — the face rows are unchanged by construction (73.9% / 61.9%) and **scene agreement falls to its null**: valence sign 4.7% vs 3.4% ($z{+}1.6$), family 18.0% vs 17.3% ($z{+}0.6$); keeping the action and posture clauses gives 15.9% vs 11.6% ($z{+}2.7$) and 21.0% vs 16.8% ($z{+}2.2$). All-emotional under the people-aware template: sign 51.7% vs 36.9% ($z{+}10.1$), family 47.1% vs 34.5% ($z{+}8.8$).

Four readings follow. (1) The image reading rests on one word — the describer's expression answer — the same conclusion the 99.8% copy rate reaches independently; in effect the pipeline is "ask the describer for the facial expression and map that word to a family". (2) The scene result in the headline table ($z{+}5.8$) was an artefact of the template asserting a person and an expression for drawings with none; under an honest description this reader finds nothing above chance in scene-classified drawings. (3) The face/scene contrast is therefore *faces, and only faces, for this reader*, not "faces more than scenes". (4) Whether scene drawings hold a feeling a person can read is exactly what the human check tests; a scene-capable describer is the main

further-work item.

9 The demo application

A small local web application runs the real pipeline live: choose a drawing from the dataset or upload an image, see the five factual answers, see both emotion readings side by side, and see the similarity scored against chance. It exists to make the method inspectable rather than to collect anything.

Every number it shows is referenced to the measured chance line: the axis starts at the metric's real floor of 41, the middle half of random pairings is bracketed, and the headline reads *closer than N% of random pairings*. Nothing a visitor types or uploads is stored, and the request log is suppressed so no visitor IP addresses are recorded.

10 Data handling and pre-processing

No human participants were involved and no new data was collected. A human-rating study was designed in full and deliberately *not* run for time; it stands as the clear next study. In its place a 120-drawing blind single-coder check has been set up (sample and coding sheets drawn; coding pending).

10.1 Name redaction

Following the supervisor's direction — redact real names, keep illness and grief content because it is not identifying on its own and is essential to the analysis, and document the procedure — a three-stage pre-processing step was built.

1. **Detect**, with deliberately high recall. About 78% of the corpus is written entirely in lower case, so capitalisation alone would miss most names. Candidates come from a 7,944-name gazetteer, from mid-sentence capitalisation checked against a ∼236,000-word dictionary, and from naming phrases such as *called X* or *my friend X*.
2. **Review.** Every candidate carries a recorded decision with a reason. Only names of private individuals are redacted; places, brands, public figures and medical eponyms such as Parkinson's and Alzheimer's are kept. The audit fails if any candidate is unreviewed.
3. **Apply**, scoped to the specific record each name was reviewed in.

Eight records contained names of private individuals; those become `[name]`. Two design lessons are recorded because both were real errors caught by the audit:

- **Unscoped redaction over-reaches.** A bare token list destroyed *Jack Frost* and *woolly mammoth*, because those strings occur innocently elsewhere.
- **Redaction can forge duplicates.** Two records holding two *different* people's full names both become `[name]`, which made the analysis count them as one prompt drawn twice. Duplicate identity is therefore decided on a key derived from the original text. Those two records consist of a name alone after redaction; they carry no analysable content and are excluded from the cleaned specificity set.

Redaction changes no top-1 emotion family and moves no GoEmotions probability by more than 0.012, so the prompt-side labels are unaffected. The effect of redaction on the downstream similarity measures was not separately verified.

Records are redacted rather than withheld, following supervisor direction, so that the analysed corpus and the demonstrator show the same 1,012 records and cannot diverge; withholding would also have removed those records from the analysis. Redaction does not eliminate re-identification risk entirely, since the drawing and the residual prompt text remain.

11 Limitations — stated plainly

1. **This measures automated agreement, not human perception.** Both readers are models, and they disagree with each other in opposite directions: the CLIP baseline skews negative on line art, while the description model reports a happy or smiling expression for 57% of the faces it can read (337 of 591; 66% of all expression answers). Only human ratings could adjudicate. This is the single most important limitation.
2. **The describer invents expressions, and the reader relays them.** The description template asserts a person and an expression for every drawing: 414 of the 421 scene-classified records report no people, 406 of those still carry an expression word, and 332 answer smile, smiling or happy. Where the expression answer names an affective state (41% of records) the image reader's top-1 family follows it 99.8% of the time, so the "describe, don't ask" separation is weaker than it sounds. Of the 38 high-confidence disagreements, 19 trace to the describer.
3. **The face/scene split is partly circular.** It is governed by the describer's people-count answer, and the emotion is read from that same description. Scenes score lower because the template asserts a person and an expression for a drawing with no person, so the image-side reading there is largely noise — not because scene descriptions lack expression content (406 of 414 have it). The face rows survive the null, the clustered test, deduplication and the ablation; the scene rows do not survive the ablation (valence sign 4.7% vs 3.4%, $z{+}1.6$ under the people-aware template), so the contrast is faces, and only faces, for this reader.
4. **The valence–arousal measure is a designed proxy**, built on hand-set emotion coordinates. It is not a continuous lexicon-based rating, and it cannot separate faces from scenes.
5. **No claim is made about colour.** The corpus is monochrome, but there is no colour condition to compare against.
6. **No positive-versus-negative direction is claimed.** An earlier claim was withdrawn: the single-valence subsets have degenerate nulls, and the two readers disagree about the direction.
7. **The specificity set is the text reader's own output.** 68 of its 405 prompts contain a plain affect word and two consist of a name alone; only one measure is informative on it.
8. **Three families are too rare for family-level claims** (fear $n = 16$, anger $n = 12$, disgust $n = 1$ on the prompt side; the reader never returns disgust). Claims are restricted to overall agreement and the face/scene contrast.
9. **One generator, with unrecorded parameters.** The repeated-prompt groups pool text typed by different visitors and re-submissions by one; model version, seed and any system prompt are unknown.
10. **No human anchor yet.** The 120-drawing single-coder check is drawn but not yet coded.

12 Reproducibility

All analysis is driven by scripts with a fixed random seed and no manual steps. Every reported number is written to `outputs/analysis_final.json` and its sibling result files (`inference.json`, `consistency.json`, `method_comparison.json`, `audits.json`, `error_taxonomy.json`); the figures (`make_figures.py`) and the numeric tests (`test_numeric.py`) read those files, so no number is typed by hand downstream and a change in the analysis fails the build. One definition of every measure lives in `emoviz/metrics.py`. The automated self-test (`test.sh`) covers the environment, data integrity, the redaction audit, every analysis script, a live run of the pipeline on fresh images, the pinned numeric values and figure regeneration.

13 What this contributes

The contribution is **not** a novel emotion classifier. It is: a question nobody had asked of real first-person self-report; a dataset that cannot be downloaded; an evaluation design with a specificity check, null baselines and clustered confidence intervals throughout; a like-for-like method comparison with a paired test; a repeated-prompt consistency analysis referenced to its own null; a template ablation showing what the image reading rests on; an error taxonomy of where the readers disagree; and a documented, tested name-redaction pre-processing step.

All figures and numbers in this document were regenerated and verified against the analysis on 9 September 2026. Authoritative results file: `outputs/RESULTS_final.md`.